Alexander Kagan

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Research Interests

My research lies broadly in the area of **statistical network analysis** and is mainly motivated by **biomedical applications**:

- Modeling shared and group latent structures in collections of networks with applications to multiomics & neuroscience
- Modeling of information diffusion on networks with applications to biological transport networks & epidemiology
- Deep learning for biomarker discovery from temporal, functional, network-linked, and/or hierarchically structured data.
- Cross-validation and model selection in network-assisted regression problems

Education

University of Michigan — *Ann Arbor, MI, USA*
Ph.D. in Statistics (co-advised by Prof. Elizaveta Levina and Prof. Ji Zhu) — *2021 - 2026*

Skolkovo Institute of Science and Technology — *Moscow, Russia*
M.S. in Computer Science — *2020 - 2021*

Yandex School of Data Analysis — *Moscow, Russia*
M.S. equivalent certificate in Data Science — *2019 - 2021*

National Research University Higher School of Economics — *Moscow, Russia*
B.S. in Mathematics (Summa Cum Laude) — *2016 - 2020*

Work Experience

Yale School of Public Health — *New Haven, CT, USA*
Postdoctoral Associate (advised by Prof. Hongyu Zhao) — *July 2026 - present*

- Prediction of drug toxicity from the molecule's chemical structure coupled with the multiomics information

Sanofi — *Cambridge, MA, USA*
R&D Computational Science Intern (advised by Prof. Ziv Bar-Joseph) — *Summer 2024*

- Developed statistical tools based on Temporal Graph Neural Networks for discovering new biomarkers governing the patient's recovery process, with applications to psoriasis and Crohn's disease

Kirshner Lab, Harvard Medical School — *Cambridge, MA, USA*
Research Assistant (advised by Prof. Leonid Peshkin) — *January 2021 - April 2024*

- Developed a universal *Kinome Regularization* framework for detecting kinases governing a given phenotype
- Built a deep-learning framework for automatic detection and annotation of cell organelles in liver images
- Led a group of 3 MSc students developing supervised hierarchical variable selection methods for bulk & single-cell data

MRM Proteomics — *Montreal, QC, Canada*
Research Intern (advised by Prof. Christoph Borchers) — *Summer 2021*

- Developed dimension reduction techniques allowing robust extraction of cancer biomarkers from patients' proteomics and metabolomics measurements

Juicy Labs — *Moscow, Russia*
Junior Data Scientist — *July 2019 - February 2020*

- Developed new credit scoring models using linear regression, random forest, and boosting

Publications

Published or Accepted

1. Kagan, A., Levina, E., Zhu, J. (2026) Flexible Modeling of Information Diffusion through a Network with Statistical Guarantees. *Accepted to JMLR*, arXiv preprint (Work received 4 poster awards in 2023)
2. Mathur, S., Kagan, A., Passaban, P., Mattoo, H., Hasanaj, E., Bar-Joseph, Z. (2026) Temporal Foundation Models for Clinical Transcriptomics Data. *Accepted to Bioinformatics*, bioRxiv preprint
3. Kagan, A., MacDonald, P., Levina, E., Zhu, J. (2026) Latent Space Models for Grouped Multiplex Networks. *Accepted to JMLR*, arXiv preprint (Honorable mention for a poster at SNAB-2024)
4. Noe, M., Parisi, E., Rifat, S., Navitskis, L., Conway, D., Deshmukh, A., Kagan, A., Millward, D., Chung, E. (2024) Comparison of 1st Year and 3rd Year ECGs in Collegiate Athletes. *Journal of the American College of Cardiology*

Under Review

5. Nano, M., Harwood, J., Kagan, A., Lukaszewicz, G., Kirschner, M., Peshkin, L., Montell, D. (2025+) A kinome inhibitor screen implicates adhesion and growth factor signaling in cellular recovery after caspase activation. *Under review at Frontiers in Pharmacology*, bioRxiv preprint

In Preparation

6. Kagan, A., Tang, T., Levina, E., Zhu, J. (2026+) Cross Validation for Network Regression. (Work presented at ICSDS-2025 and JSM-2026)
7. Kagan, A., Rata, S., Gruver, J. S., Trikoz, N., Lukyanov, A., Vultaggio, J., Ceribelli, M., Thomas, C., Gujral, T. S., Kirschner, M. W., Peshkin, L. (2026+) An Optimal Set of Inhibitors for Reverse Engineering via Kinase Regularization.
8. Kriukov, D., Efimov, E., Kagan, A., Peshkin, L. (2026+) On Universal Approach to Biological Age Estimation

Software

1. `InfluenceDiffusion` (2024)
   PYPI Python package for estimation and uncertainty quantification of influence diffusion models on networks
2. `GroupMultiNeSS` (2025)
   PYPI Python package for extraction of shared and group latent structures from a collection of multiplex networks

Presentations

Contributed Talks

2025	**ICSDS (International Conference on Statistics and Data Science)**, Seville, Spain
2024	**CMStatistics (Conference on Computational and Methodological Statistics)**, London, UK
2024	**JSM (Joint Statistical Meetings)**, Portland, OR, USA
2023	**JSM (Joint Statistical Meetings)**, Toronto, ON, Canada

Posters

2024	**SNAB workshop (***Honorable mention***)**, Nassau, Bahamas
2023	**Michael Woodroofe Memorial Conference** *(Honorable mention)*, Ann Arbor, MI, USA
2023	**SNAB workshop** *(Best poster award)*, Anchorage, AK, USA
2023	**ICSA Applied Statistics Symposium** *(Honorable mention)*, Ann Arbor, MI, USA
2023	**MSSISS Statistics Symposium** *(Best poster award)*, Ann Arbor, MI, USA

Teaching Experience

GRADUATE STUDENT INSTRUCTOR, UNIVERSITY OF MICHIGAN

- **Stats 601: Advanced Statistical Learning** (Ph.D. level) *Winter 2026*
  Developed and taught weekly labs (~30 students), held office hours, graded homework, and exams
  For teaching this course, I received a Graduate Student Instructor Excellence in Teaching Award
- **Data Science 415: Data Mining and Statistical Learning** (upper undergraduate level) *Fall 2025*
  Taught and prepared Python notebooks for lab sections (~20 students), graded homework and exams
- **Stats 485: Capstone Seminar** (upper undergraduate level) *Fall 2022*
  Held office hours, graded data analysis reports.
- **Stats 250: Introduction to Statistics and Data Analysis** (lower undergraduate level) *Winter 2022*
  Taught weekly lab sections (~40 students), held office hours, graded homework and exams.
- **Stats 426: Introduction to Theoretical Statistics** (upper undergraduate level) *Fall 2021*
  Held office hours, graded homework and exams.

TEACHING ASSISTANT, HIGHER SCHOOL OF ECONOMICS, MOSCOW, RUSSIA

- **Machine Learning** (upper undergraduate level) *Winter 2020*
  Designed and taught weekly lab sections(~30 students), graded homeworks.
- **Real Analysis** (lower undergraduate level) *2018 - 2019*
  Held office hours, graded homeworks and exams.

Awards

- Graduate Student Instructor Excellence in Teaching Award (awarded by UM) *2026*
- Outstanding Graduate Student Service Award (awarded by UM) *2025*
- Rackham Travel Grant (awarded by UM) *2022-2025*

Professional Service

- **Organizing Committee Chair of MSSISS Statistics Symposium** *2024-2025*
  Received an Outstanding Graduate Student Service Award for chairing the organizing committee of MSSISS, an annual statistics symposium for ~150 people at UMichigan. My responsibilities included: student and keynote speaker invitations, funding requests, catering, website maintenance, and booklet printing.
- **Levina-Zhu Research Group Admin** *2022-2025*
  Responsible for the logistics of my advisors' research group: scheduled weekly reading group presentations, curated group's Google Drive, website, and networks literature database.
- **Organizing Volunteer at ICSA Statistics Symposium** *June 2023*
  Helped participants with conference logistics and registration.

Computing Skills

Proficient in Python (Numpy, Pandas, Sklearn, Matplotlib, PyTorch, Scipy, NetworkX, JAX, CVXPY), R, and Matlab

Languages

English (fluent), Russian (native), German (upper-intermediate), French (intermediate), Hebrew (intermediate)